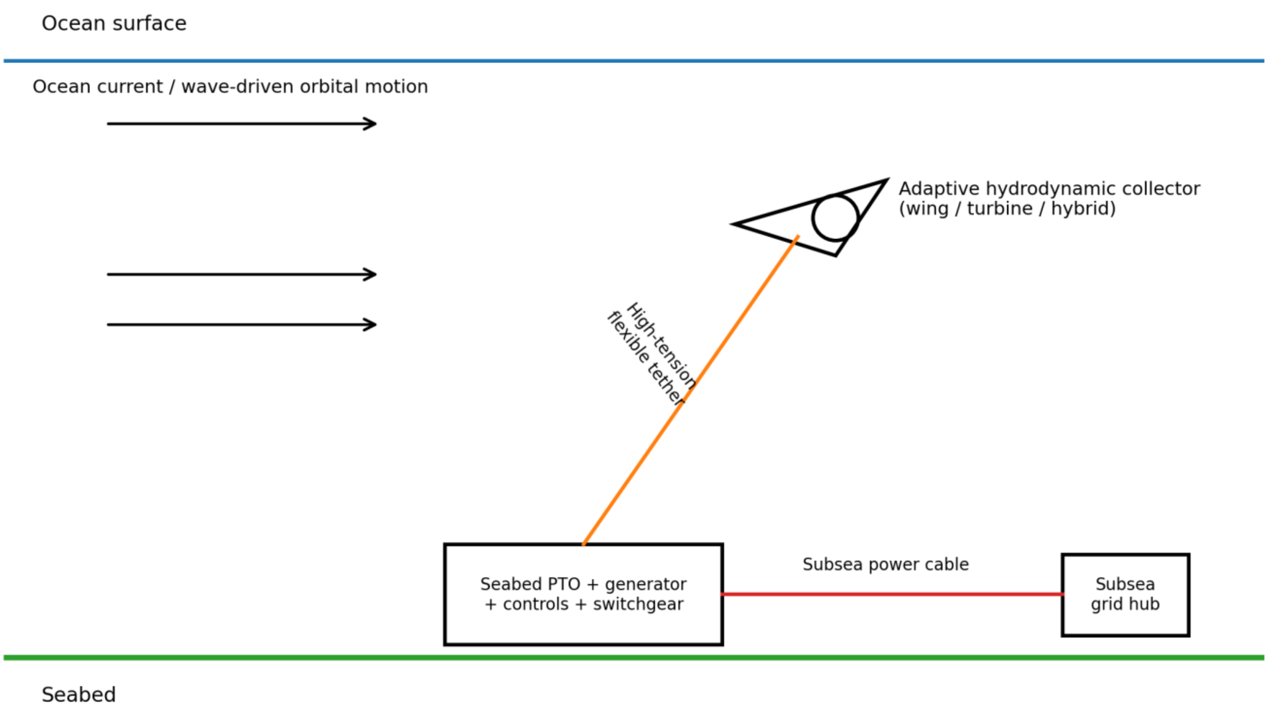
SEABED-ANCHORED ADAPTIVE OCEAN ENERGY NETWORK

A concept white paper for high-force, high-swept-volume marine energy conversion using tethered collectors and seabed grid nodes



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Purpose: Open technical contribution and structured evaluation. This document does not claim demonstrated performance or completed patent novelty.

Abstract

This white paper proposes a distributed marine-energy architecture in which permanent or semi-permanent power conversion and grid infrastructure is placed on or anchored to the seabed, while comparatively lightweight hydrodynamic collectors move through the water on controlled flexible tethers. The central hypothesis is that a fixed seabed reaction structure, combined with cross-current or pendular collector motion, can convert a modest net force imbalance into high collector velocity, large swept water volume, and high cyclic tether work. Mechanical or hydraulic motion rectification, gearing, accumulators, flywheels, and power electronics would convert irregular bidirectional motion into grid-compatible electrical output.

The paper distinguishes the concept from ordinary floating wave devices, fixed tidal turbines, and existing underwater kites. Several underlying elements are established in current marine-energy research: tethered underwater wings, cross-current trajectories, seabed anchoring, winch-based power take-off, and subsea electrical export. The proposed contribution is the integrated system architecture: a seabed power-and-grid node supporting one or more adaptive collectors, with explicit use of high reaction force, controlled tether displacement, multi-mode energy capture, motion rectification, and networked subsea aggregation.

A device-level output increase approaching 100× relative to a narrowly defined passive reference device is shown to be mathematically possible under favorable assumptions involving relative velocity, effective swept area, and conversion efficiency. It is not established as an expected field result. The paper therefore defines a staged validation program based on first-principles energy accounting, coupled hydrodynamic simulation, bench-scale power-take-off testing, and progressively larger water trials.

Executive proposition

> **Keep the heavy, durable, grid-forming infrastructure fixed at the seabed; allow only the energy-harvesting collector to move. Use the seabed as a reaction structure, the tether as a controllable force-and-motion path, and the subsea network as a scalable power system.**

Key claims and non-claims

Claim / hypothesis	Status in this paper
A fixed seabed reaction structure can sustain high tether loads.	Established engineering principle; site- and foundation-dependent.
Cross-current collector motion can raise relative flow speed.	Established in underwater-kite research and prototypes.
Large tether force plus controlled displacement can produce useful power.	Direct consequence of $P = Fv$; requires dynamic motion, not static tension.
Gears, one-way clutches, hydraulics, or power electronics can rectify irregular motion.	Established component technology; does not create energy.
The integrated architecture can outperform passive floating collectors by orders of magnitude.	Testable hypothesis; reference case and equal-resource boundary must be defined.

1. Problem statement and motivation

The ocean contains kinetic and potential energy in waves, tides, persistent currents, density-driven circulation, and localized high-flow corridors. Conventional marine-energy devices typically choose one of two structural strategies: remain near the surface and move with the sea, or fix a turbine relative to the seabed and allow water to pass through it. The proposed system explores a third strategy: fix the power conversion base and grid node to the seabed, but deliberately allow the collector to execute a large controlled trajectory through the water.

1.1 Design intuition

A freely floating body reduces relative motion by moving with the surrounding water. A fixed reaction point permits tension to develop. If a lightweight collector is attached to a high-capacity seabed foundation, a controlled difference between hydrodynamic driving force and generator/tether resistance can accelerate the collector. The collector may then traverse the current crosswise, sweep a much larger water volume than its own frontal area, or execute repeated pendular arcs. The machine is therefore conceived as a dynamic energy harvester rather than a static turbine.

1.2 Why the seabed matters

- It provides a stable reaction structure for large cyclic tether loads.
- It can house heavy power-take-off, generator, hydraulic, grid, and protection equipment away from the most violent surface environment.
- It can support a network topology in which multiple collectors share power conversion, switchgear, communications, and export infrastructure.
- It permits the moving collector to be lighter, recoverable, and replaceable while leaving the long-life foundation and grid node in place.

1.3 Important physical caution

The holding capacity of an anchor or foundation is not an energy source. Large static force alone produces no power. Continuous power requires force through distance over time. The design objective is therefore not maximum tension by itself, but maximum sustainable cycle-averaged work after hydrodynamic, mechanical, electrical, and availability losses.

$$P = F \cdot v \quad \text{and} \quad E = \int F \cdot ds$$

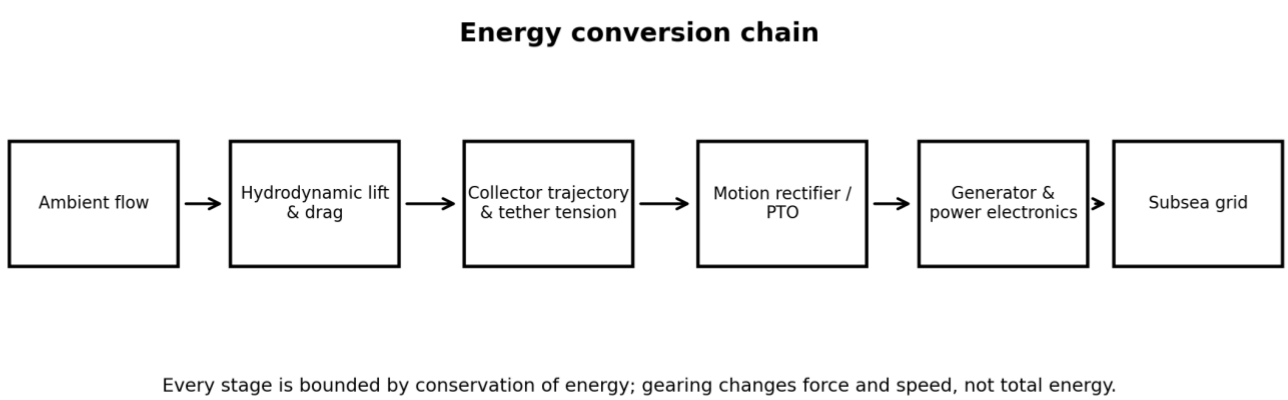

Figure 1. Energy conversion chain and conservation boundary.

2. Proposed system architecture

2.1 Seabed foundation and grid node

Each node comprises a foundation, structural interface, power-take-off (PTO), generator or hydraulic motor-generator, power electronics, protection devices, sensing, communications, and a subsea cable connection. Foundation choice may include gravity base, suction caisson, driven pile, drilled anchor, rock anchor, or multi-point mooring depending on bathymetry and geotechnical conditions.

2.2 Adaptive hydrodynamic collector

The collector is a buoyancy-controlled wing, turbine-wing hybrid, ducted rotor, oscillating foil, or multi-surface body. It may align passively with local flow, or use control surfaces and variable buoyancy to seek a favorable depth and execute a commanded trajectory. The collector is intentionally lighter and more replaceable than the seabed node.

2.3 High-tension tether

The tether transmits structural loads and may also transmit electrical power, hydraulic pressure, mechanical traction, data, or a combination. Candidate constructions include synthetic fiber with armored conductors, segmented umbilicals, separate load and power members, or a mechanical tether paired with an independent low-strain cable. Bending fatigue, torsion, abrasion, marine growth, and termination design are critical.

2.4 Power-take-off alternatives

- Traction winch with reel-out generation and low-energy reel-in recovery.
- Bidirectional drum with dual one-way clutches or mechanical motion rectifier.
- Hydraulic cylinder or rotary pump feeding a high-pressure accumulator and steady-speed hydraulic generator.
- Direct-drive linear generator located inside the seabed node.
- Onboard turbine-generator with electrical export through the tether, while tether tension provides trajectory control.
- Hybrid mode: onboard turbine plus base-mounted traction generation.

2.5 Subsea collection network

Nodes connect to local subsea collection hubs. Depending on scale and distance, the network may use medium-voltage AC, medium-voltage DC, or high-voltage DC export. The architecture should support isolation of failed nodes, black-start assistance from shore or storage, and modular addition of new collectors without redesigning the entire farm.

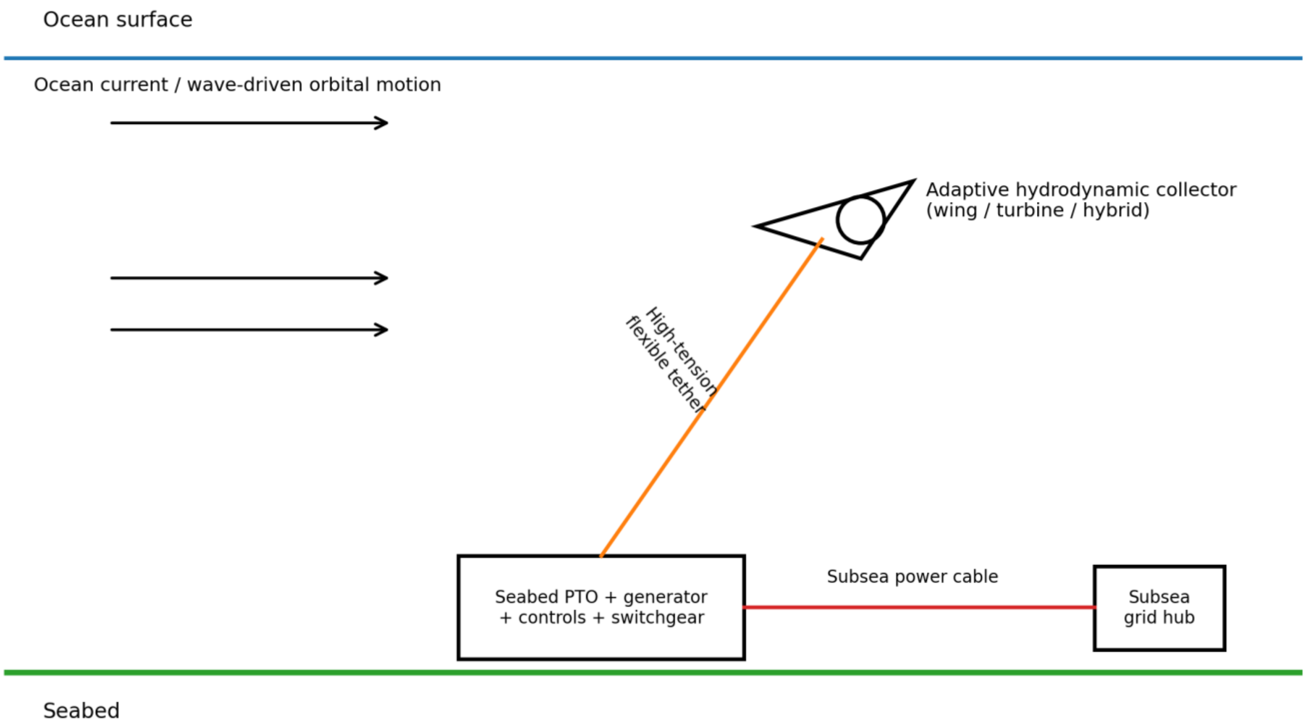

Figure 2. Conceptual seabed node, tethered collector, and subsea grid connection.

3. Operating modes

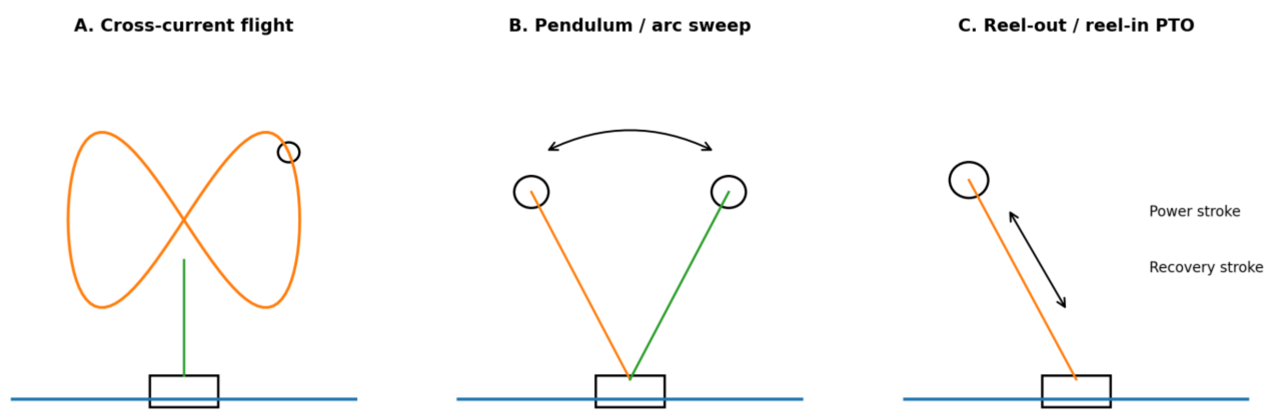

Figure 3. Candidate operating modes. A final system may combine more than one mode.

3.1 Cross-current flight

A lifting wing flies transverse to the ambient current, often in a figure-eight path. Its motion can create a relative flow across the turbine that is several times the ambient stream speed. This principle is already recognized in underwater-kite systems. The proposed architecture extends it by considering a common seabed power node, alternative base-mounted PTO, multiple collectors, and multi-mode operation.

3.2 Pendulum or arc sweep

Wave orbital motion, changing currents, buoyancy, and tether geometry drive the collector through a controlled arc. The PTO resists motion during energy-producing portions of the cycle and reduces resistance during recovery. The trajectory may be passive, semi-passive, or actively controlled.

3.3 Reel-out / reel-in generation

The collector generates high tether tension while moving outward. The tether drives a generator through a drum, gearbox, or hydraulic system. During recovery, the collector changes orientation to reduce hydrodynamic force and reels in using a fraction of the generated energy. The net cycle energy is the power-stroke output minus recovery energy and losses.

3.4 Multi-collector orchestration

Two or more collectors can be phased so that one produces while another recovers, smoothing output and improving utilization of the seabed generator. A common accumulator, flywheel, DC link, or battery can further reduce short-term fluctuations. Collision avoidance and hydrodynamic wake interaction become central control problems.

4. First-principles physics

4.1 Hydrodynamic force

A simplified hydrodynamic force scale is:

$$F\_h \approx ½\, \rho\, C\, A\, V\_rel^2$$

where ρ is water density, C is an effective lift or drag coefficient, A is reference area, and V_rel is collector velocity relative to the local water. Because force scales with the square of relative velocity, cross-current motion can create much larger forces than a static body in the same ambient current.

4.2 Flow power and turbine power

$$P\_flow = ½\, \rho\, A\, V\_rel^3$$

The cubic velocity relationship explains the attraction of cross-current motion. However, V_rel is not an independent free input: it results from a coupled balance among hydrodynamic lift, collector drag, turbine loading, tether drag, gravity, buoyancy, trajectory curvature, and control forces.

4.3 Collector acceleration from force imbalance

$$m\_c\, a = F\_drive - F\_tether - F\_drag - F\_PTO - F\_other$$

The user’s core hypothesis is captured here. A high-capacity seabed reaction structure allows large tether force, while a comparatively small residual force may accelerate a lighter collector. The foundation mass does not multiply energy; it prevents the reaction point from following the collector. The achievable trajectory is then governed by the net force and the collector mass.

4.4 Tether power

$$P\_tether = T \cdot v\_reel$$

T is tether tension and v_reel is the speed of controlled tether motion at the PTO. Very high tension can produce high power even at modest reel speed. If the tether is locked and v_reel = 0, traction power is zero, although an onboard turbine could still generate.

4.5 Gears and force-speed trade-off

An ideal gearbox conserves mechanical power apart from losses: increasing torque decreases speed proportionally. One-way clutches, mechanical rectifiers, hydraulic check valves, accumulators, flywheels, and controlled electrical converters can turn oscillatory input into smoother unidirectional output, but cannot exceed the mechanical work delivered by the ocean to the collector.

$$P\_out = \eta\_total \cdot P\_mechanical, \quad with\ 0 < \eta\_total < 1$$

5. The 100× hypothesis: a disciplined formulation

The phrase "100× more energy" is meaningful only after defining the reference system and equalizing boundaries. A comparison against a small passive floating rotor of equal rotor area is different from a comparison against a mature offshore wind farm, a fixed tidal turbine using equal swept area, or an optimized underwater kite. The proposed concept should therefore report multiple ratios rather than one universal multiplier.

5.1 Velocity-area-efficiency ratio

$$R\_P \approx R\_A \cdot R\_V^3 \cdot R\_\eta$$

where R_A is the ratio of effective intercepted or swept area, R_V is the ratio of relative flow speed at the energy conversion surface, and R_η is the ratio of net conversion efficiencies.

Scenario	Area ratio	Velocity ratio	Efficiency ratio	Power ratio
Conservative	1.2	2.0	0.65	6.2×
Strong	1.5	3.0	0.7	28.4×
100× design point	2.0	4.0	0.78	99.8×
Aggressive	2.0	4.6	0.7	136×

Table 1. Illustrative scaling only; these are assumptions, not measured performance.

5.2 Force-motion ratio

$$R\_P = R\_F \cdot R\_v \cdot R\_\eta$$

If an anchored architecture delivers 1,000× the useful force of a passive reference but only 0.15× its conversion-path velocity, and has 70% of the reference efficiency, then the idealized ratio is 105×. The difficulty is proving that the 1,000× force comparison is physically fair and sustainable over repeated displacement. Anchor holding capacity alone is not equivalent to hydrodynamic driving force.

5.3 Energy-flux boundary

No device can continuously extract more energy than is supplied to its control volume by incoming flow, waves, pressure fields, and replenishment. A collector that sweeps a larger path may access a larger control volume, but neighboring devices and downstream flow will interact. Therefore, a 100× device-level advantage may coexist with a much smaller farm-area advantage.

5.4 Recommended claim language

> **Hypothesis: the combination of a fixed seabed reaction structure, high-tension controlled tether motion, cross-current velocity amplification, enlarged effective swept volume, and efficient motion rectification may enable order-of-magnitude device-level gains relative to passive floating collectors. A 100× design point is mathematically admissible under favorable assumptions and requires coupled simulation and experimental validation.**

6. Comparison with existing offshore technologies

Attribute	Floating wave device	Fixed tidal turbine	Underwater kite	Floating offshore wind	Proposed network
Reaction reference	Mooring / inertia	Seabed structure	Seabed tether	Mooring / buoyancy	Seabed PTO/grid node
Moving mass	Platform/absorber	Rotor	Wing + turbine	Large platform + tower	Light collector; base fixed
Primary gain mechanism	Wave-relative motion	Local current through rotor	Cross-current speed	Large wind rotor	Cross-current speed + tether work + swept volume
PTO location	Surface / internal	Nacelle underwater	Often onboard or tether-linked	Nacelle	Preferably seabed; hybrid possible
Output smoothing	Hydraulics/electronics	Rotor inertia/electronics	Trajectory/control	Rotor/grid controls	Phased collectors + accumulator/DC link
Maturity	Prototype to early commercial	Early commercial	Demonstration/early deployment	Commercial / scaling	Concept
Primary risk	Survivability	Underwater maintenance	Control + tether	Capital cost / moorings	Tether fatigue + subsea maintenance + control

Table 2. Qualitative comparison. Existing underwater-kite systems are the closest prior engineering class.

6.1 Relation to existing underwater kites

The U.S. Department of Energy defines tidal kites as hydrodynamic wings with attached turbines tethered to fixed points; looping motion increases speed around the turbine and permits extraction in slower currents. Minesto publicly describes a figure-eight underwater kite whose turbine experiences flow several times the ambient stream speed. SeaQurrent publicly describes a system with a seabed mooring anchor, tether, and hydraulic energy converter. These references establish that important parts of the proposed concept are technically credible and already active areas of development.

6.2 Distinguishing system-level emphasis

- Permanent seabed node as shared mechanical, electrical, control, and grid infrastructure.
- Explicit use of controlled high tether force and force imbalance to shape collector velocity.
- Optional hybrid extraction: onboard turbine plus seabed traction or hydraulic PTO.
- Multiple collectors per node with phase-shifted trajectories for output smoothing.
- Multi-mode operation across current-dominant, wave-dominant, and mixed conditions.

7. Mechanical and electrical design concepts

7.1 Non-reversing mechanical conversion

A bidirectional input can be converted into unidirectional generator rotation using dual one-way clutches, differential gear trains, mechanical motion rectifiers, or hydraulic check-valve circuits. The preferred approach should minimize backlash, shock loading, lubrication demands, and inaccessible wear surfaces. In deep-water service, simplicity and sealed modules may be more valuable than peak laboratory efficiency.

7.2 Hydraulic PTO

Hydraulics are attractive for high force and low speed. A tether drum or cylinder can pressurize fluid during both directions of motion through rectifying valves. A gas-charged accumulator smooths pulsating input, and a hydraulic motor drives a generator at near-constant speed. Risks include seal life, fluid leakage, pressure cycling, and efficiency at partial load.

7.3 Direct electrical PTO

A permanent-magnet linear generator or low-speed rotary generator can reduce moving parts. Power electronics can rectify variable-frequency generation into a DC link. The trade-off is increased generator mass and the need for pressure-resistant insulation and connectors.

7.4 Control architecture

- Local fast loop: collector attitude, tether tension, reel speed, and trajectory tracking.
- Node supervisory loop: maximize cycle-averaged net power while respecting fatigue and foundation limits.
- Farm controller: phase collectors, avoid collisions, reduce wake interference, manage export constraints.
- Fail-safe modes: feather, neutral buoyancy, controlled parking, tether release, or seabed docking.

7.5 Instrumentation

- Tension, strain, bend radius, and acoustic emission sensors in tether and terminations.
- ADCP/current profiler, pressure, wave spectrum, turbulence, and temperature sensors.
- IMU, depth, heading, velocity, and proximity sensors on collector.
- Generator torque/speed, hydraulic pressure/flow, DC-link power, insulation monitoring, and leakage detection.

8. Engineering constraints and failure modes

Risk	Mechanism	Potential consequence	Design response
Tether fatigue	Millions of tension and bending cycles	Progressive damage or rupture	Separate load/power paths; bend limiters; replaceable tether cartridge; continuous health monitoring
Tether twist	Changing trajectory and yaw	Conductor damage; control instability	Swivels, torque-balanced tether, trajectory constraints
Foundation overload	Extreme current, storm, resonance	Seabed failure or structural damage	Load shedding, feathering, sacrificial release, geotechnical safety factors
Collector runaway	Control or actuator failure	Overspeed, collision, tether shock	Passive stable geometry, redundant control, emergency drag/feather surfaces
Biofouling	Marine growth on wing, rotor, sensors	Reduced lift and efficiency	Coatings, cleaning cycle, replaceable skins, performance monitoring
Subsea electronics failure	Pressure, leakage, insulation degradation	Node outage	Oil-filled/pressure-compensated modules, redundancy, wet-mate connectors
Maintenance cost	Specialized vessels and weather windows	Poor economics	Towable collector, ROV-friendly interfaces, modular retrieval, shallow first deployments
Ecological interaction	Collision, noise, entanglement, habitat change	Permitting and biodiversity impact	Slow rotor tips, exclusion controls, acoustic monitoring, adaptive curtailment

Table 3. Principal technical risks and initial mitigation concepts.

8.1 Depth strategy

The first prototype should not target the abyssal ocean. Continental-shelf or channel sites with manageable depth, known current resource, accessible ports, and existing subsea capabilities are preferable. The active collector can operate in the most energetic water layer while the foundation remains at the seabed. Very long tethers increase drag, control delay, catenary complexity, and recovery difficulty.

8.2 Storm survival

The system should distinguish operating mode from survival mode. During extremes, the collector may feather, descend to a calmer depth, dock to the seabed node, reduce projected area, or release through a recoverable safety mechanism. Designing only for peak energy extraction would be unsafe; survivability and fatigue life are first-order economic variables.

9. Modelling and validation framework

9.1 Stage 0 — analytical model

1. Define the reference technology and comparison boundary: equal rotor area, equal collector mass, equal material mass, equal occupied ocean area, or equal installed cost.
2. Build a lumped-parameter model for lift, drag, tether tension, reel speed, turbine load, buoyancy, and trajectory curvature.
3. Compute cycle-averaged mechanical work, recovery energy, generator losses, and availability assumptions.
4. Perform sensitivity analysis for current speed, tether length, wing area, mass, control law, and PTO damping.

9.2 Stage 1 — coupled numerical simulation

- Unsteady CFD or validated hydrodynamic coefficients for wing/turbine interaction.
- Six-degree-of-freedom multibody dynamics with flexible tether model.
- PTO, gearbox/hydraulic, generator, and power-electronic model.
- Optimal control or model-predictive control for maximum net energy subject to load limits.
- Irregular waves, turbulent currents, shear profile, direction changes, and fault scenarios.

9.3 Stage 2 — laboratory water channel

Build a small wing or collector with instrumented tether and controllable PTO emulator. Measure lift, trajectory, tension, reel speed, net cycle work, and stability. Test passive alignment, controlled figure-eight flight, pendulum mode, and reel-out/reel-in mode. The objective is not grid power but validation of the coupled dynamic model.

9.4 Stage 3 — sheltered-water prototype

Deploy a 1–10 kW class prototype at moderate depth with a recoverable seabed frame. Validate autonomous operation, fail-safe recovery, marine growth effects, tether life, and real net energy. Use modular wet-mate connectors and a simple shore cable.

9.5 Stage 4 — open-sea demonstrator

Scale to 100 kW–1 MW only after the model accurately predicts smaller tests. Demonstrate seasonal energy yield, maintenance hours per MWh, survival through design sea states, environmental monitoring, and grid compliance.

10. Performance metrics and decision gates

Metric	Definition	Why it matters
Net cycle efficiency	Electrical energy exported / hydrodynamic-mechanical energy delivered to collector	Prevents gross force or gross power from masking recovery and conversion losses
Device power ratio	Net average power / reference device net average power	Tests 10×–100× hypotheses with a declared reference
Farm-area power density	Annual MWh / occupied km^2	Captures spacing, wakes, and resource depletion
Specific power	Average kW / tonne of installed system	Tests material efficiency
Capacity factor	Annual average output / rated output	Measures utilization and predictability
Availability	Time able to operate / total time	Subsea reliability is essential
Maintenance intensity	Vessel hours or intervention cost / MWh	Often dominates marine-energy economics

Tether damage equivalent load	Fatigue-equivalent cyclic load	Key life-limiting variable
LCOE	Discounted lifetime cost / lifetime electricity	Ultimate commercial metric

10.1 Suggested decision gates

- Gate A: model predicts >10× device-level gain against a clearly defined passive reference without violating energy-flux limits.
- Gate B: water-channel tests reproduce trajectory and net-work predictions within ±15%.
- Gate C: sheltered-water prototype achieves positive net cycle energy with autonomous recovery and no critical tether damage.
- Gate D: open-sea demonstrator shows competitive specific power and credible maintenance pathway.
- Gate E: array simulation shows farm-level advantage after spacing, wakes, export losses, and availability.

11. Potential Indian relevance

India has a long coastline, significant offshore engineering capability, growing electricity demand, island and remote-load applications, and strategic interest in domestic clean-energy technologies. The concept should initially be evaluated as an ocean-engineering research program rather than a near-term replacement for nuclear, solar, wind, or conventional hydropower.

11.1 Candidate institutional pathway

- Ministry of New and Renewable Energy (MNRE): renewable-energy R&D and policy alignment.
- Ministry of Earth Sciences (MoES) and National Institute of Ocean Technology (NIOT): ocean resource, materials, seabed systems, testing, and marine operations.
- IIT ocean engineering, naval architecture, mechanical engineering, controls, and power-electronics groups: modelling and prototype design.
- Public-sector offshore and subsea engineering organizations: foundations, cables, installation, and maintainability.

11.2 Recommended submission framing

A government or research submission should avoid presenting the 100× factor as a conclusion. The appropriate framing is a high-upside hypothesis with a low-cost first validation step. The request should be for independent hydrodynamic and system modelling, not immediate large-scale deployment.

> **Proposed evaluation question: Can a seabed-anchored adaptive collector, using controlled cross-current motion and high-tension tether power conversion, achieve an order-of-magnitude increase in net power-to-material ratio relative to passive marine-energy devices while maintaining acceptable fatigue life and maintenance cost?**

11.3 Open-contribution posture

The author's intent is to contribute the concept for legitimate technical evaluation and potential public benefit. A dated white paper provides authorship and provenance without asserting exclusive ownership. Any public release should invite criticism, request transparent assumptions, and welcome independent replication.

12. Research questions

5. What is the optimal collector mass, buoyancy, wing loading, and tether length for a given current-speed distribution?
6. How much relative velocity amplification is achievable after turbine loading and tether drag are included?
7. Can a common seabed PTO efficiently serve multiple phased collectors?

8. Which conversion path—onboard turbine, base traction PTO, hydraulic PTO, direct electrical PTO, or hybrid—maximizes lifecycle energy?
9. What is the highest sustainable tether-force-to-moving-mass ratio under realistic fatigue limits?
10. Can a collector passively seek stronger current layers, or is active depth and trajectory control necessary?
11. What survival configuration minimizes extreme loads without expensive retrieval?
12. What is the device-level gain versus a passive reference, and what is the farm-area gain versus mature alternatives?
13. How do wakes, resource replenishment, and array spacing limit large-scale deployment?
14. Can maintenance be designed around recoverable collectors and replaceable tether/PTO cartridges while the foundation remains in place?

13. Conclusion

The seabed-anchored adaptive ocean energy network is a plausible system-level concept grounded in established physical principles and adjacent marine-energy technologies. Its central design move is to separate the durable reaction and grid infrastructure from the dynamic collector. A large-capacity seabed foundation permits controlled tether loading; a lighter collector can convert force imbalance into trajectory and relative velocity; the PTO converts high tension through displacement into electrical power; and a subsea grid aggregates many modular nodes.

The concept does not obtain energy from anchor weight or gearing. It obtains energy from ocean flow and wave fields. The foundation enables reaction, the collector enlarges accessible motion and relative flow, and the conversion system attempts to capture a larger fraction of the available energy. A 100× device-level result is not ruled out by simple scaling equations when velocity amplification, swept-area gain, and efficiency are combined, but it remains an unvalidated design point. The most important next step is a transparent coupled model that compares equal boundaries and reports net cycle energy, not peak force.

The idea is therefore suitable for disciplined scrutiny. Its value lies not in a guaranteed multiplier but in a potentially powerful architecture that can be tested incrementally, falsified honestly, and improved through ocean engineering, controls, materials, power electronics, and subsea systems research.

References and technical context

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Appendix A — Example preliminary calculation template

The following template is intended for a future spreadsheet or simulation. Values must be site-specific and experimentally validated.

Input	Symbol	Illustrative value / note
Water density	ρ	1025 kg/m^3
Ambient current speed	U	Site distribution, not one value
Collector relative speed	V_{rel}	Output of dynamic model
Wing/collector reference area	A_w	Design variable
Turbine area	A_t	Design variable
Lift/drag coefficients	C_L, C_D	Function of angle, Reynolds number, fouling
Collector mass and buoyancy	m, B	Include added mass
Tether length and diameter	L_t, d_t	Include drag and elasticity
Tether tension	$T(t)$	Dynamic output
Reel speed	$v_{reel}(t)$	Control variable
PTO efficiency	η_{PTO}	Load-dependent
Generator/electrical efficiency	η_e	Load-dependent
Recovery energy	E_{rec}	Subtract from gross output
Availability	A_v	Include weather and maintenance

Appendix B — Proposed simulation outputs

- Collector 3D trajectory and relative velocity distribution.
- Tether tension, angle, curvature, and fatigue cycle histogram.
- Instantaneous and cycle-averaged hydrodynamic power, PTO power, generator power, and exported power.
- Energy produced during power stroke and consumed during recovery.
- Constraint violations: maximum tension, minimum depth, collision envelope, generator torque, foundation load.
- Sensitivity plots for current speed, turbulence, wave spectrum, tether length, wing area, mass, and control parameters.
- Comparative ratios against declared reference technologies under identical resource conditions.
- Array-level wake, spacing, cable loss, availability, and farm-area power density.